LoQANT User Manual

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Content:

Overview

LoQANT (Localisation and Quantification of Antigen Nuclear Translocation) is a plugin for ImageJ/Fiji. It is designed to assess the nuclear positivity of monitored antigens in both H-DAB and fluorescence-stained samples. It is particularly useful for monitoring the translocation of nuclear receptors and other molecules from the cytoplasm to the nucleus. A major advantage of this script is that it only evaluates the positive signal in the nucleus, independent of cytoplasmic staining. In addition to displaying nucleus-positive cells, the staining intensity in the nuclei can also be measured semiquantitatively or quantitatively, depending on the staining method.

Other advantages of the provided algorithm include: elimination of inter-observer visual perception bias; minimal supervision required for analysis, thus significantly eliminating the heavy reliance on a trained pathologist/histologist; reduction of time required for analysis; only a few steps are required to follow the analysis and measure staining intensity.

General recommendation

When using LoQANT, it is recommended to follow good image analysis practice to obtain valuable results, such as fixation, staining, image acquisition settings, exposure, contrast and brightness being consistent across a cohort of samples (Cromey, 2010). For LoQANT to work properly, you need to specify the size of the nuclei in pixels. This will filter out artefacts or overlapping cores. It is also advisable to test which of the automatic thresholding algorithms are suitable for the measured samples.

Cromey DW. Avoiding twisted pixels: ethical guidelines for the appropriate use and manipulation of scientific digital images. Sci Eng Ethics. 2010;16(4):639-67.

1 Installation

1.1 Requirements

- **Operating system:** LoQANT works on Windows, Mac and Linux operating systems (tested on Windows 10, Windows 11, macOS Monterey 12.5., and Ubuntu 22.04 LTS operating systems)
- **Software:** The latest version of FIJI (last tested on version 2.14.0/1.54f)
- **Image formats:** the image format depends on type of method:
  - H-DAB: one RGB image, white background
  - Fluorescence: two RGB or grayscale images of separate channels, black background
  - Preprocessed images: H-DAB: two grayscale images of separate channels, white background

1.2 Download Fiji

Download the latest version of FIJI at https://imagej.net/software/fiji/downloads according to your operating system and install it.

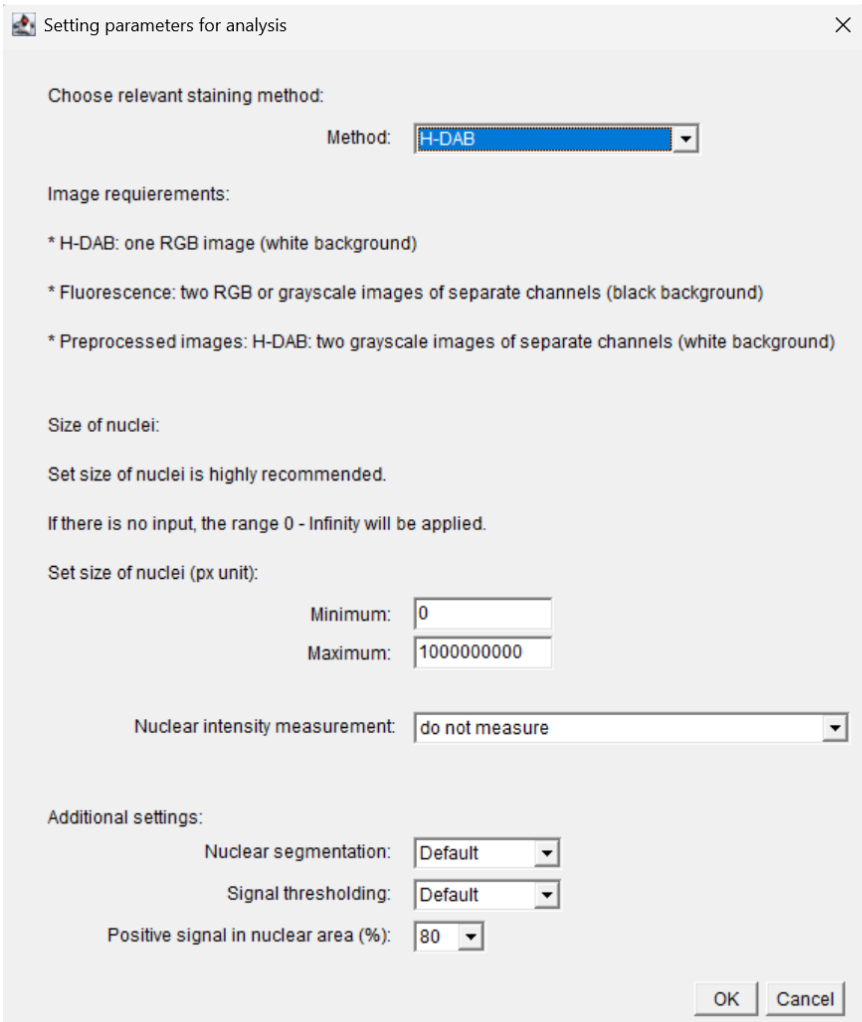
1.3 Download LoQANT

LoQANT is freely available on GitHub at following link: https://github.com/Keri-histo/LoQANT

1. Download the LoQANT_.py file.
2. Find the Fiji.app folder on your computer.
3. Open the Fiji.app folder and place LoQANT_.py file to folder called „plugins“.
4. Launch Fiji and LoQANT is now to be found in Plugins menu. (Note: If the Fiji has been already open, it needs to be restarted.)

2 Running LoQANT

1. Open Fiji and open image or images for analysis. The image (images) must be open before launching LoQANT.
2. In the plugins menu, select LoQANT. You will find it near the bottom of the list. This will open the initial dialog box (see below).



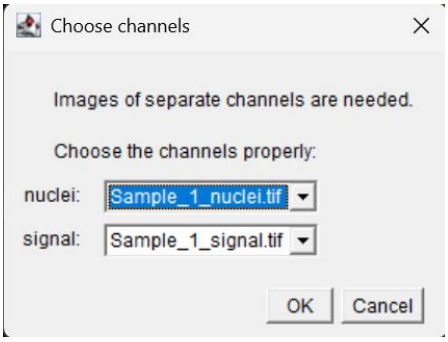
3. Choose the staining method used for your sample from the menu. You can select from three options: „H-DAB“, „fluorescence“ or „preprocessed images: H-DAB“.
4. Specify the range of nuclei sizes (in pixel units) to be included in the analysis. The size specification is used to filter out particles that are too small (which are usually artefacts) and also clusters of cells where the analysis is distorted by the overlay. If you do not specify the nuclei size, the default setting of „0 – Infinity“ will be applied.
5. Choose if you want to measure the intensity of the nuclear staining. By default, no measurement is set („do not measure“ option). In this case, LoQANT will only analyse the percentage of positive cells. There are three intensity measurement options to choose from:
   a. A semiquantitative measurement, which is suitable for DAB-stained samples (choose th „semiquantitative (rec for DAB)“ option). This method measures the number and percentage of the positive nuclei in the following categories: weak, medium, and high intensity. It then calculates the histoscore and returns the overall positivity category for a given field of view.

   b. A quantitative measurement of intensity value for each positive nuclei („quantitative: positive nuclei (rec for fluorescence)“ option). This method measures the intensity of staining as the mean gray value in nuclei considered as positive. Quantitative measurement is recommended for fluorescent samples.

   c. The quantitative measurement of all detected nuclei („quantitative: all nuclei (rec for fluorescence)“ option). It measures intensity of staining as the mean gray value in all detected nuclei, regardless of whether they are considered positive or not. Quantitative measurement is recommended for fluorescent samples.

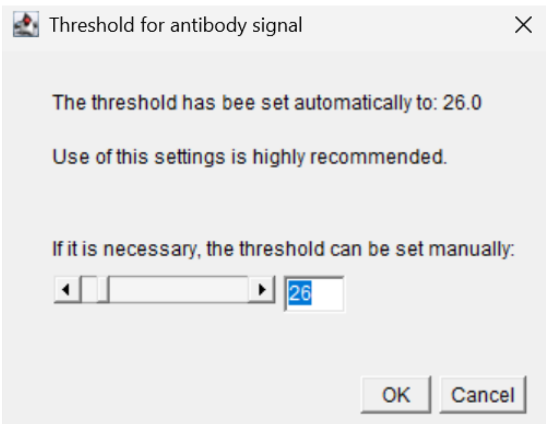
6. In the Additional Settings section, you can select the algorithm for segmenting the nuclei and the positive signal. You can use all of the AutoThreshold algorithms available in Fiji (Default, Huang, Huang2, Intermodes, IsoData, Li, MaxEntropy, Mean, MinError, Minimum, Moments, Otsu, Percentile, RenyiEntropy, Shanbhag, Triangle and Yen).You can also set the percentage of nuclei that must have a positive antibody signal for all nuclei to be considered positive.

7. If you select either the „fluorescence“ or „preprocessed image“ method, the next dialog box will ask you to specify which of the opened images is for nuclei and antibody signal. All images opened in Fiji will be listed in the menu.



8. The nuclei and antibody signal will be segmented automatically using AutoThreshold algorithms that you selected in initial dialog.

9. If the antibody thresholding does not work as expected, you can change the threshold value using the slider. The automatically set threshold value will be displayed in the upper right corner of the dialog box. After confirming the threshold value, the plugin will evaluate the image (see below).



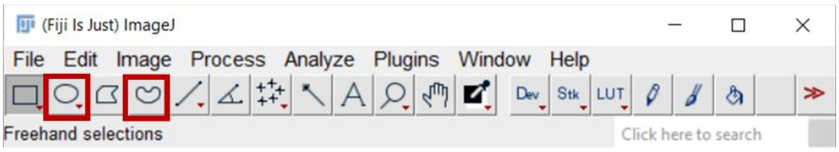
IMPORTANT NOTICE: The ROI Manager must stay open during whole the analysis.

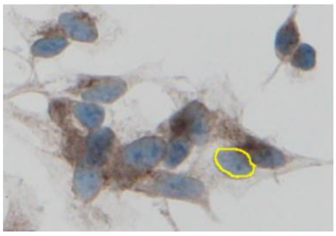
3 Input parameters determination

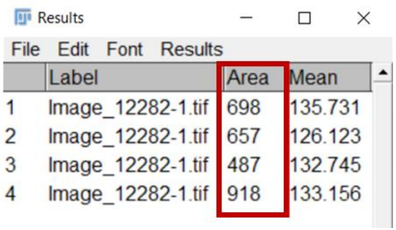
3.1 How to measure the size of a nucleus

1. LoQANT requires the size in pixel units. So, before you start, check units of measurement: select "Set Scale" in the "Analyze" menu and check in the corresponding dialog box. If you've set any units, click "Click to Remove Scale" to remove them.

2. To measure the size of the nuclei in the analysed sample, select either "oval" or "freehand selection" in the toolbar and mark the nucleus. You can then measure its size either via the "Analyze - Measure" menu or by pressing the "m" key. The Result Table will open, the size is in the Area column.






Results
File Edit Font Results

	Label	Area	Mean
1	Image_12282-1.tif	698	135.731
2	Image_12282-1.tif	657	126.123
3	Image_12282-1.tif	487	132.745
4	Image_12282-1.tif	918	133.156

3.2 Selecting the AutoThreshold algorithm

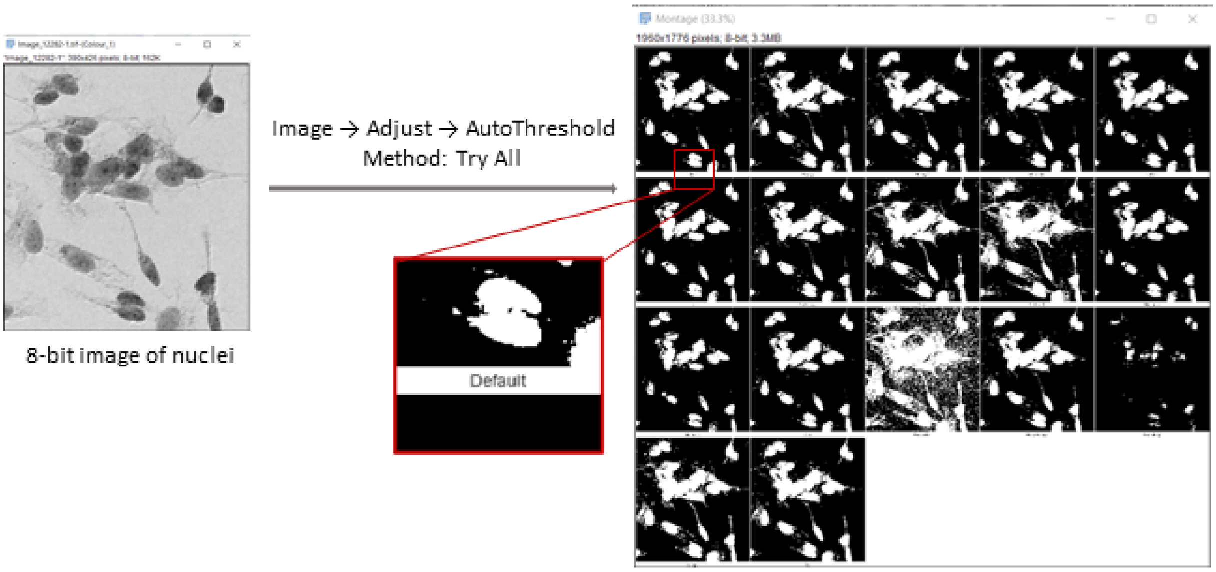
1. You can use all the algorithms for automatic thresholding available in AutoThreshold (ver. 1.18.0), which is part of Fiji.

2. To select the appropriate algorithm for nuclei segmentation, you will need an 8-bit image representing the nuclei.

3. If the staining is fluorescence, the channel is separate and just needs to be converted to 8-bit (if it is not already) via the "Image - Type - 8-bit" menu.

4. If the staining is DAB, you need to deconvolve the colours first. This is done via the menu "Image - Color - Color Deconvolution". After deconvolution the image is 8-bit.

5. Navigate menu "Image - Adjust - AutoThreshold" and select "Method: Try all". It is also necessary to check whether the objects are white objects on a black background (in the case of fluorescence) or not (in the case of DAB). After confirmation, the segmentation result is displayed using all 14 available algorithms. The name of the algorithm used is below each image. Choose the one most suitable for your sample.



4 Test images

You can test LoQANT function using provided test images.

H-DAB.tif

- RGB image
- open this image in Fiji
- select method: H-DAB
- set nuclei size to: 300-1500

fluorescence_nuclei.tif and **fluorescence_signal.tif**

- RGB images of separate fluorescence channels
- open both images in Fiji
- select method: fluorescence
- set nuclei size to: 100-800
- in following dialog box, set:
  - fluorescene_nuclei as image represents nuclei
  - fluorescence_signal as image represents signal

preproccessed_haematoxylin_nuclei.tif and **preproccessed_DAB_signal.tif**

- 8-bit grayscale images generated as result of color deconvolution of H-DAB image
- open both images in Fiji
- select method: fluorescence
- set nuclei size to: 100-800
- in following dialog box, set:
  - preproccessed_haematoxylin_nuclei as image represents nuclei
  - preproccessed_DAB_signal as image represents signal